

Supplementary Material

1 SUPPLEMENTARY TABLES

Supplementary tables are provided as Supplementary_Tables_CTGFHT.xlsx and as individual CSV files. Figure source-data tables are provided in the source_data folder.

2 SUPPLEMENTARY FIGURES

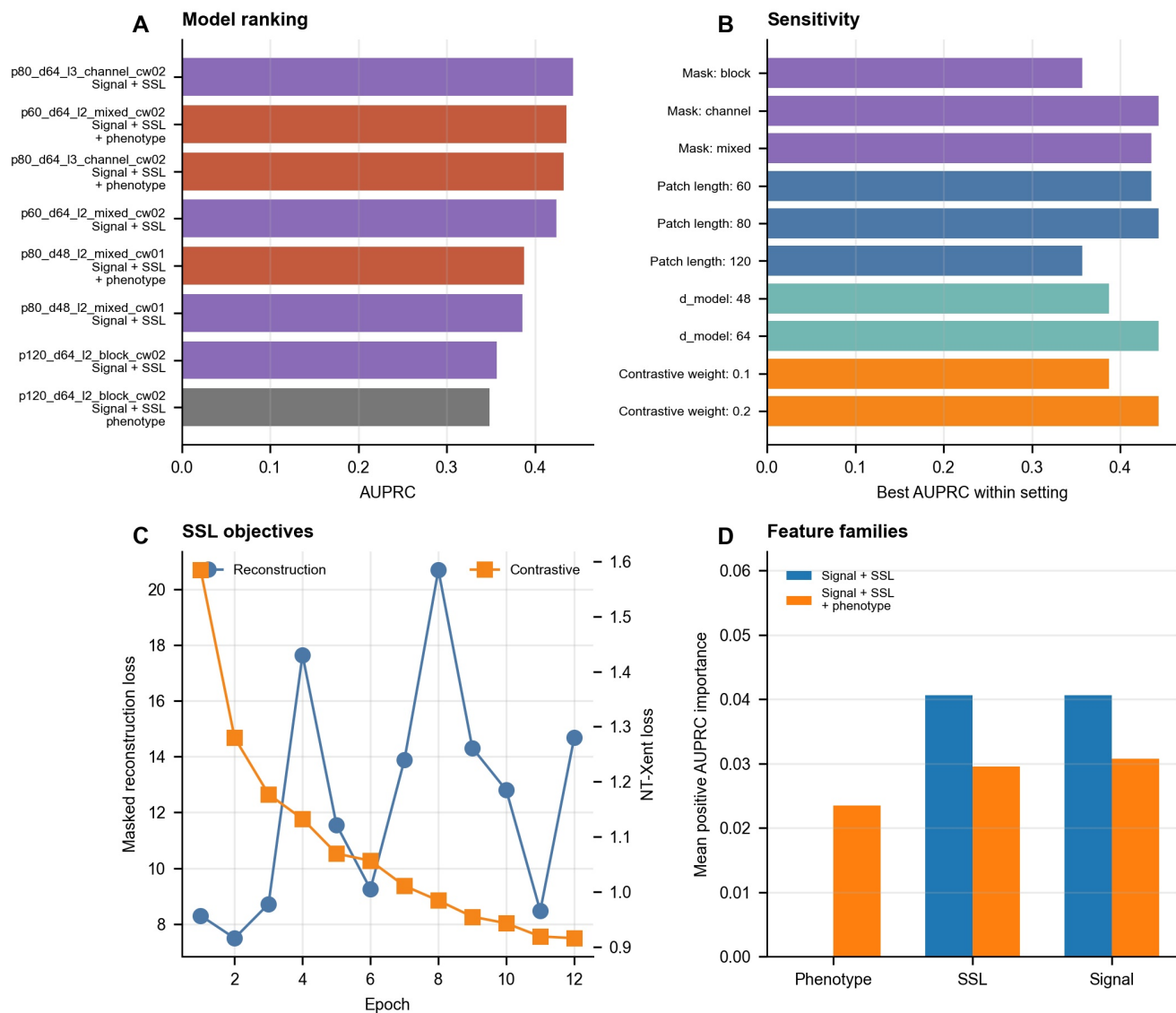


Figure S1. AI model-selection and sensitivity panels for the transformer phenotyping workflow. Supplementary analyses of the self-supervised AI method layer. (A) Exploratory transformer configuration and feature-set combinations ranked by held-out AUPRC. (B) Sensitivity summary across masking strategy, patch length, model dimension and contrastive objective weight, using the best AUPRC observed within each setting. Because these panels use held-out performance for method exploration, they should not be interpreted as a locked independent model-selection experiment. (C) Training-objective trace for a selected-architecture refit, showing masked reconstruction loss and NT-Xent contrastive loss across epochs. (D) Feature-family contribution to AUPRC-based permutation importance for signal-plus-SSL and final phenotype-augmented models.

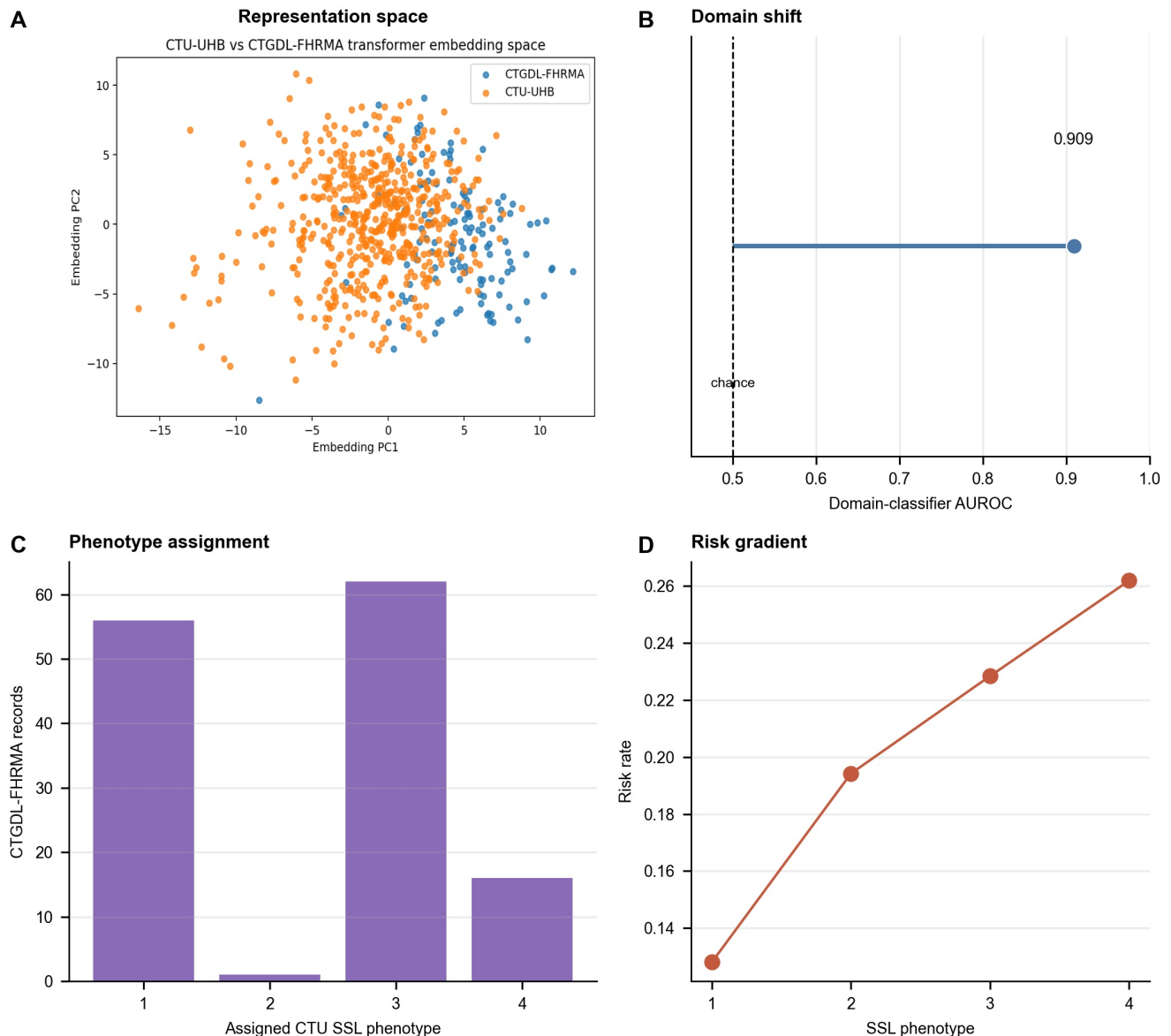


Figure S2. CTGDL-FHRMA supports external representation and domain-shift validation. External representation analysis using CTGDL-FHRMA records. Because external neonatal outcome labels were not used for CTGDL-FHRMA, this analysis is interpreted as external representation/domain-shift validation rather than external outcome validation. (A) Principal-component visualization of CTU-UHB and CTGDL-FHRMA transformer embeddings. (B) Domain separability quantified by a CTU-versus-CTGDL classifier. (C) CTGDL-FHRMA records assigned to CTU-derived SSL phenotypes. (D) CTU-UHB neonatal risk gradient across SSL phenotypes.